



Optical computing to produce AI

Few startups are using optical computing to build laser-powered systems that work much faster than standard chips. <http://dgit.in/LightComp>



Hawking's multiverse

Stephen Hawking's final paper, A Smooth Exit from External Inflation, sought to find proof for the multiverse theory. <http://dgit.in/HawkLegacy>

Bacteria vs Cancer

According to a recent study, a particular strain of skin bacteria can protect humans against skin cancer.

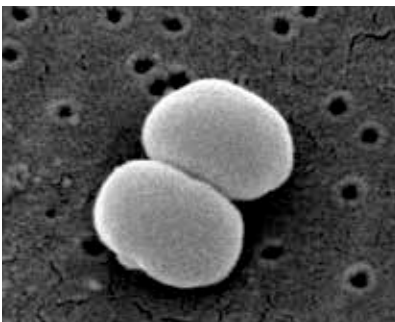
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hile people mostly perceive bacteria as something bad, it is mostly known to the scientifically inclined

and health conscious that there are several good bacteria as well. Highlighting the importance of such bacteria, a new study reveals that a particular strain of bacteria commonly found on the human body could be useful in fighting skin cancer.

A particular strain of *Staphylococcus epidermidis* was discovered by researchers at the University of California, San Diego, to have the selective ability to inhibit the growth of certain forms of cancer. Interestingly, this discovery was serendipitous.



Scanning Electron Micrograph of *Staphylococcus epidermidis*

The researchers were studying strains of the *Staphylococcus* species because they are known to exhibit the ability to inhibit harmful bacteria. Their objective was to distinguish the

anti-microbial properties of the different strains - essentially how this bacteria behaved against other microorganisms. While conducting this research, Prof Richard Gallo, a co-author of the research along with his team found something interesting about *Staphylococcus epidermidis*. In its ability to inhibit infections such as strep throat, it also produced a substance similar to adenine, a key component of DNA.

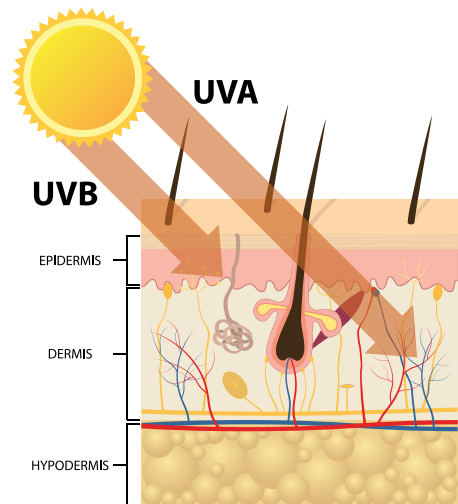
Once it was identified, the chemical was tested to determine if it would act against tumours. 6-N-hydroxyaminopurine (6-HAP) was found to successfully hinder the production of DNA and hence exhibit the ability to act against several types of tumors. In terms of safety, it was also found to be non-toxic. In experiments with mice, the test subjects who were injected with it showed tumors 60% smaller than the ones in the mice who did not receive the injections.

This does not essentially imply that people without this strain are more vulnerable to skin cancer, or that everyone with this strain will experience its ability to defend against the tumours. According to the team, while *Staphylococcus epidermidis* is quite commonly found on human skin, about only 20% of the healthy population is likely to have a strain which produces 6-HAP. As Prof. Gallo puts it, the strain is "common, but not on everyone". *Staphylococcus epidermidis* colonizes moist body surfaces such as the axillae, nares and toe webs. So now you know where to look for them.

Skin cancer is one of the most commonly occurring forms of cancer around the world. Considering that the ultraviolet radiation found in sunlight is one of the primary causes behind skin cancer, it is also one of the most unconsciously acquired forms as well. With further re-

search on this discovery, the team at University of California expects the discovery of newer ways to deal with skin cancer even before it affects someone. Think of it like a vaccination but against cancer.

While the impact of this discovery is significant, it is not something new for bacteria residing in the human body to come to the aid of its hosts.



Ultraviolet radiation from sunlight causes skin cancer

A lot of our primary functions, especially in the digestive system, depend on the microbiomes that are found in almost every part of our body. For instance, *Lactobacillus acidophilus* is an intestinal bacteria that helps in the digestion process by preventing the formation of harmful bacteria. Vegans should be particularly thankful to *Bacteroides Thetaiotaomicron* as they help in breaking down quite a few plant molecules. Another example is the *Viridans Streptococci* that acts as a resilient tenant in the throat that leaves little space for other, more harmful bacteria to proliferate.

So maybe the next time you take a bath or hit the handwash, maybe give these good guys a second thought? (We're kidding, don't do that). 